



EE712 - Embedded Systems Design

Electrical Engineering IIT Bombay

Laboratory Experiment - II

Problem set: 2.2

Date: January 23, 2025

DAC Interfacing

Aim: To get familiarized with the MCP4921 and SSI protocol to send data from TM4C123GH6PM to an external DAC.

Components to be used:

- TIVA C Board
- Potentiometer
- DAC (MCP4921)

Problem Definition: Given a 1-K Ω potentiometer, connect it to PE3 port of the TIVA C Board as you have performed in labsheet 3. Swipe the voltage between 0 to 3.3V and convert it to digital signal using on-chip ADC and send the ADC data to external DATA using SSI protocol.

General Instructions

- Find the DAC datasheet here : MCP4921 Datasheet.
- The DAC uses SPI interface. Include "driverlib/ssi.h" in your .c file.
- Enable one of the SSI modules using the function SysCtlPeriphEnable(), Page 505.

Pin Name	Pin Number	Pin Mux / Pin Assignment	Pin Type	Buffer Type ^a	Description
SSI0Clk	19	PA2 (2)	I/O	TTL	SSI module 0 clock
SSI0Fss	20	PA3 (2)	I/O	TTL	SSI module 0 frame signal
SSI0Rx	21	PA4 (2)	I	TTL	SSI module 0 receive
SSI0Tx	22	PA5 (2)	O	TTL	SSI module 0 transmit

Figure 1: SSI Pins to be used

- Set the pin type of the associated pins as SSI. Use GPIOPinTypeSSI(), Page 281 for this.
- Configure the SSI with appropriate parameters. Use SSIConfigSetExpClk(), Page 469 for this.
- Use SysCtlClockGet(), Page 488 for system clock.
- Enable the SSI using the function SSIEnable(), Page 473.
- Set up the ADC as you did in labsheet 3.
- Set up the DAC as instructed in the below given figure.
- Construct the appropriate frame for DAC as given below:

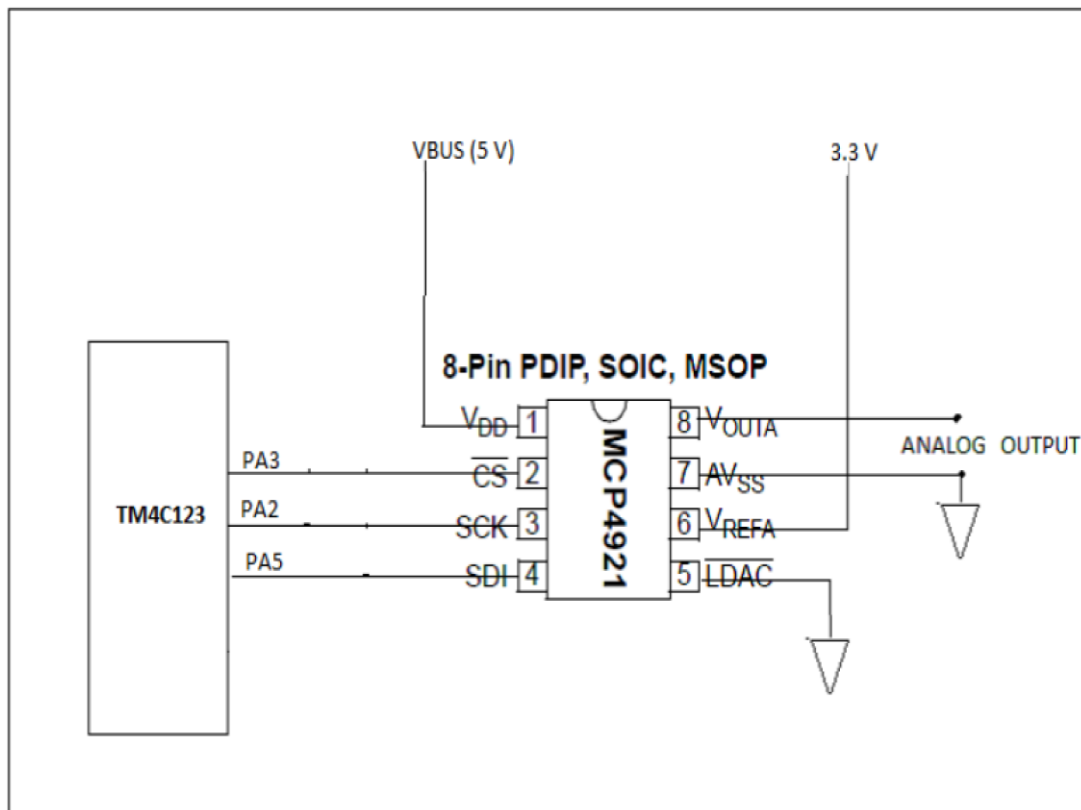


Figure 2: Interfacing MCP4921 with TM4C123

bit 15	$\overline{A/B}$: DAC _A or DAC _B Select bit 1 = Write to DAC _B 0 = Write to DAC _A
bit 14	BUF: V _{REF} Input Buffer Control bit 1 = Buffered 0 = Unbuffered
bit 13	$\overline{GA}$: Output Gain Select bit 1 = 1x ($V_{OUT} = V_{REF} * D/4096$) 0 = 2x ($V_{OUT} = 2 * V_{REF} * D/4096$)
bit 12	$\overline{SHDN}$: Output Power Down Control bit 1 = Output Power Down Control bit 0 = Output buffer disabled, Output is high impedance
bit 11-0	D11:D0: DAC Data bits 12 bit number "D" which sets the output value. Contains a value between 0 and 4095.

Figure 3: DAC frame to be sent from TM4C123 to MCP4921

- Send the frame to DAC by SSIDataPut(), Page 471. Wait till SSI is busy. Check using SSIBusy(), Page 467.
- Measure the voltage at the output pin of DAC and check it whether it is equal to the voltage given to ADC.
- Make sure you have included the following header files: "stdint.h", "stdbool.h", "inc/hw memmap.h", "inc/hw ssi.h", "inc/hw types.h", "driverlib/gpio.h", "driverlib/sysctl.h", "driverlib/pin map.h", "driverlib/adc.h" and "driverlib/ssi.h".
- Make sure the entire waveform you give to the ADC is POSITIVE. Provide appropriate DC bias to ensure that the input is in range.
- This DAC accepts 16-bit digital data as input. Make sure you pad the ADC data with proper control bits before sending it to DAC.

Console configuration:

Go through **Page 5-13 to 5-15** to configure the variable to display it in the CCS consle of the following link.
TM4C123G_LaunchPad_Workshop_Workbook

References:

1. TM4C123GH6PM : TM4C123GH6PM_Datasheet
2. TIVA Driver Library: TIVA_Driver_Library
3. TM4C123G LaunchPad Workshop Workbook TM4C123G_LaunchPad_Workshop_Workbook
4. MCP4921_Datasheet: MCP4921 Datasheet.